Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/30/2006
PI Name:	Zamorano, Lucia M.D.		
Project Title:	The Development and Human Factors Analysis of Advanced 3D-Visualization for Telepresence Applications		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	lzamorano@aol.com	**Fax:**	FY 313-9660368
PI Organization Type:	UNIVERSITY	**Phone:**	313-966-0342
Organization Name:	Wayne State University		
PI Address 1:	Neurosurgery Department		
PI Address 2:	4160 John R Suite 930		
PI Web Page:	http://www.luciazamorano.com/		
City:	Detroit	**State:**	MI
Zip Code:	48201	**Congressional District:**	13
Comments:	Co-Investigators: A. Pandya, J. Maida, M. Siadat, J. Gong, Q. Li.		
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-01
Start Date:	01/01/2000	**End Date:**	01/01/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This research is focused on the issues of Human Performance related to NASA's Critical Roadmap: "related technological research and development to enhance performance, decrease errors in critical tasks and improve human/system interface." Specifically, we address critical question 6.09 (risk type III and priority I) which asks, "What information systems, resource management methods and tools, and communications systems architecture(s) will best support the crew's ability to operate autonomously ?" Autonomous human task performance for remote operations often depends entirely on visual feedback that is indirect. Visualization of the remote data and the human performance impacts of this visualization are of utmost importance to NASA and, as it turns out, also to the medical community. Because of the risky environment of space, teleoperations are commonplace for NASA. As telepresence technology matures and these systems become more dexterous, have less time lag issues by becoming more autonomous, future satellite repair missions or station building missions could be controlled from teleoperated/telepresence sites. In this research, we use a medically orientated Neurosurgery		

	telepresence system as a test bed for the Human Factors Analysis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There has been significant technology transfer between the collaborating NASA lab (Johnson Space Center/GRAF) and the neurosurgery team at Wayne State University. The potential benefit of this technological development could be enormous not only to NASA, but also to the outcome of neurosurgerical and other surgical patients. The field of medical robotics has the highest potential benefit.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)